

3.1.2 Monotonicity Worksheet

1. Show that the sequence $a_n = \frac{2}{n}$ is strictly decreasing.
2. Determine whether the sequence $a_n = \frac{3^n}{n!}$ is eventually increasing or decreasing.
3. Show that the sequence $a_n = \frac{n}{n+2}$ is strictly increasing using the derivative/calculus technique.

4. Show how the Monotone Sequence Theorem can be used to show that the sequence $\{a_n\}_{n=2}^{\infty}$, with general term $a_n = \frac{2n+3}{n-1}$, converges.